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基於基因體基礎模型之總體基因體短讀序列分類學分類

Token-level Genomic Foundation Model Fine-tuning for Metagenomic Taxonomic Classification

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摘要

總體基因體學的分類學分類——即將每條定序短讀序列指派至正確的微生物分類單元——是微生物群落分析的核心計算挑戰。本論文系統性評估基因體基礎模型於總體基因體分類任務的適用性，並透過資料規模、預訓練、tokenization 等多重消融實驗，分析其主要瓶頸。我們提出一種基於 Nucleotide Transformer v2（498M 參數）的 Token 層級分類方法，結合低秩適應（LoRA）微調與注意力池化分類頭，保留完整的 Token 層級序列資訊。透過系統性的資料規模實驗（500K 至 50M 條平衡讀序列），我們發現在固定 NT-v2 架構下，訓練資料量是決定分類準確度的主要因素：在屬（genus）層級 120 類分類任務中，資料量由 500K 增至 5M 可提升準確度 7.76 個百分點（55.29% 提升至 63.05%），進一步擴至 50M 可達 67.07%，而架構與訓練技巧的調整最多僅帶來 ±0.5 個百分點的變化。反向互補測試時增強（RC TTA）在所有設定下穩定提供 0.1–1.5 個百分點的準確度改善。受控消融實驗證實預訓練 29 層骨幹網路比相同資料量下的單層隨機初始化模型高出 13.19 個百分點，驗證了預訓練表示在固定 tokenization 下的價值。然而，跨架構的 tokenization 消融實驗進一步顯示，在相同 50M 資料量下，採用重疊 13-mer tokenization 的 MetaTransformer 從頭訓練即可達到 87.42% 的屬層級準確度，較 NT-v2 + LoRA（67.07%）高出 20.35 個百分點，顯示 tokenization 設計是比預訓練更具決定性的因素，而 NT-v2 的非重疊 6-mer tokenizer 是其主要結構性限制。在物種（species）層級 1,535 類任務上，per-genus 50M LoRA 分類器於 oracle genus routing 下達到 27.8% Top-1，超越 sp_v4 平面 oracle 上限（27.4%），驗證階層式架構方向的正確性。樣本層級評估顯示，在物種平衡的隨機分割合成樣本中，67% 的單讀序列準確度可轉化為相對豐度估計 Pearson 相關係數 $r = 0.993$ 與豐度 $\geq$1% 之屬的 100% 偵測靈敏度，但此結果是在固定群落組成下的穩定性測試，於真實多樣性社群上的泛化仍待驗證。本研究為基因體基礎模型在總體基因體學分類上的應用提供了實證基礎與可擴展的實驗框架，並指出未來方向應聚焦於以重疊長 k-mer tokenization 預訓練的微生物基礎模型。

關鍵字：

基因體基礎模型、總體基因體學、分類學分類、低秩適應、注意力池化

Token-level Genomic Foundation Model Fine-tuning for Metagenomic Taxonomic Classification

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ABSTRACT

Metagenomic taxonomic classification—assigning each sequencing read to its source organism in the microbial taxonomy—is a fundamental computational challenge in microbiome analysis. This thesis systematically evaluates the suitability of genomic foundation models (GFMs) for metagenomic read classification through controlled ablations across data scaling, pre-training, and tokenization. We propose a token-level classification approach leveraging the pre-trained Nucleotide Transformer v2 (NT-v2, 498M parameters) with Low-Rank Adaptation (LoRA) fine-tuning and an attention-pooling head that preserves the full token-level sequence information.

Within the fixed NT-v2 framework, training data volume is the dominant factor determining classification accuracy: a $10\times$ increase in training data yields a $+7.76$ percentage point (pp) improvement in genus-level accuracy (55.29% to 63.05% on 120 genera at 5M reads, scaling further to 67.07% at 50M reads), whereas architectural and training-technique ablations produce at most ± 0.5 pp of variation. Reverse-complement test-time augmentation consistently improves accuracy by $+0.1$ to $+1.5$ pp across all experimental settings. Balanced subsampling across species reduces the number of unclassifiable taxa (F1$=$0 classes) from 9 to 0 at the 50M scale and improves macro F1 by $+5.08$ pp over the 5M baseline. A controlled backbone ablation (29-layer pre-trained NT-v2 vs. a single-layer randomly initialized Transformer at the same 50M scale and identical 6-mer tokenization) further isolates a $+13.19$ pp pre-training advantage, confirming the value of GFM pre-training under fixed tokenization.

However, cross-architecture ablations reveal that tokenization design is an even larger determinant of performance: trained from scratch on the same 50M reads, MetaTransformer with overlapping 13-mer tokenization reaches 87.42% genus Top-1, outperforming NT-v2 + LoRA (67.07%) by $+20.35$ pp. NT-v2's non-overlapping 6-mer tokenization is therefore identified as the principal structural limitation of the proposed approach, rather than backbone capacity or adaptation strategy. At the species level (1,535 classes), per-genus 50M LoRA classifiers under oracle-genus routing reach 27.8% Top-1, exceeding the flat sp_v4 oracle ceiling of 27.4% and validating the hierarchical-classification direction. In synthetic species-balanced random-partition samples, per-read errors largely cancel at the sample level (Pearson $r = 0.993$, 100% detection sensitivity at $\geq$1% abundance); evaluation on real and compositionally variable communities remains future work. This work provides empirical evidence for the relative roles of data scale, pre-training, and tokenization in GFM-based metagenomic classification, and points to overlapping long-k microbial pre-training as the most promising future direction.

Keywords:

Genomic Foundation Model, Metagenomics, Taxonomic Classification, LoRA, Attention Pooling

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Abbreviations

AMP	Automatic Mixed Precision
ART	Artificial Read Technology (read simulator)
BPE	Byte Pair Encoding
DNA	Deoxyribonucleic Acid
ESM	Evolutionary Scale Modeling
FFN	Feed-Forward Network
GFM	Genomic Foundation Model
GPU	Graphics Processing Unit
HGR	Human Gastrointestinal Reference
LA	Logit Adjustment
LCA	Lowest Common Ancestor
LoRA	Low-Rank Adaptation
LR	Learning Rate
MLM	Masked Language Modeling
MLP	Multi-Layer Perceptron
NT	Nucleotide Transformer
PEFT	Parameter-Efficient Fine-Tuning
RC	Reverse Complement
RC TTA	Reverse-Complement Test-Time Augmentation
RoPE	Rotary Position Embedding
SLURM	Simple Linux Utility for Resource Management
TTA	Test-Time Augmentation
UMGS	Unified Human Gastrointestinal Genome collection
VRAM	Video Random Access Memory

Symbols

r	LoRA rank
α	LoRA scaling factor
K	Number of classes
T	Sequence length (number of tokens)
d	Hidden dimension of the backbone model
M	Number of learnable query tokens in attention pooling
W_Q, W_K, W_V	Query, key, and value projection matrices
A, B	LoRA low-rank decomposition matrices
τ	Temperature parameter for logit adjustment
π_k	Prior probability of class k
f_θ	Classifier parameterized by θ
$\bar{r}$	Reverse complement of read r

Chapter 1

Introduction

1.1 Background and Motivation

Metagenomics, the culture-independent study of microbial communities through direct sequencing of environmental DNA, has transformed our understanding of microbial ecology, human health, and environmental science [1]. Modern metagenomic sequencing platforms, particularly Illumina short-read technologies, generate hundreds of millions of reads per sample, each typically 100–300 base pairs (bp) in length. A fundamental computational challenge in metagenomic analysis is *taxonomic classification*: assigning each sequencing read to the correct taxon in the microbial taxonomy, such as genus or species [2].

Accurate taxonomic classification underpins virtually all downstream metagenomic analyses, including community composition profiling, pathogen detection, antibiotic resistance gene tracking, and comparative microbiome studies [3]. However, the classification task is inherently difficult due to several factors:

- **Short read length**: Individual reads of 150 bp carry limited phylogenetic signal, especially when distinguishing closely related taxa.
- **High taxonomic diversity**: A typical metagenomic sample may contain reads from hundreds to thousands of species spanning dozens of genera.
- **Class imbalance**: Community compositions follow highly skewed abundance distributions, with a few dominant taxa and many rare ones.

- **Sequence similarity**: Closely related taxa share substantial genomic homology, making fine-grained discrimination (e.g., species-level) particularly challenging.

Traditional approaches to metagenomic classification fall broadly into two categories. *Alignment-based methods* such as BLAST [4] and DIAMOND [5] align reads against reference databases, achieving high precision but at considerable computational cost. *k-mer-based methods* such as Kraken2 [6] and Centrifuge [7] index reference genomes using k-mer hash tables, enabling ultra-fast classification at the expense of sensitivity. More recently, deep learning approaches have emerged, including MetaTransformer [8], which applies a Transformer architecture to learned k-mer embeddings for metagenomic read classification.

Concurrently, the field of genomics has witnessed the development of *genomic foundation models* (GFMs)—large-scale pre-trained models that learn general-purpose representations of DNA sequences from billions of nucleotides [9]. Models such as the Nucleotide Transformer [9], DNABERT [10], DNABERT-2 [11], and Evo [12] have demonstrated strong performance across diverse genomic tasks, including promoter prediction, splice site identification, and variant effect prediction. These models offer a compelling alternative to task-specific feature engineering: by pre-training on vast genomic corpora, they capture rich sequence semantics that can be transferred to downstream tasks through fine-tuning.

Despite their promise, the application of GFMs to metagenomic taxonomic classification remains underexplored, and several open questions remain. A naïve approach —extracting mean-pooled embeddings from a frozen pre-trained model and training a linear classifier—discards the positional and local information encoded in individual token representations. A separate but equally important question concerns *tokenization*: most existing GFMs (including NT-v2) use non-overlapping or short-k tokenization schemes inherited from masked-language-model pre-training, yet k-mer-based metagenomic classifiers commonly rely on overlapping long k-mers (e.g., $k = 12$–13) for taxonomic specificity. It is therefore unclear whether large-scale GFM pre-training can compensate for tokenization choices that are mismatched to the downstream taxonomic task.

This thesis investigates two complementary questions: (i) whether a *token-level* approach, preserving the full sequence of token embeddings and employing parameter-efficient fine-tuning, can substantially improve metagenomic taxonomic classification over frozen-embedding baselines; and (ii) whether GFM pre-training is sufficient to compensate for the absence of task-specific k-mer tokenization, or whether tokenization design imposes a structural ceiling that pre-training alone cannot overcome.

1.2 Problem Statement

Given a set of metagenomic short reads $\{r_1, r_2, \ldots, r_N\}$ generated by sequencing a microbial community, the taxonomic classification problem is to assign each read r_i to its correct taxon $y_i \in \{1, 2, \ldots, K\}$, where K is the number of classes at a given taxonomic level (e.g., genus or species).

This thesis formulates the problem as a supervised multi-class classification task. We assume access to a reference genome database from which simulated reads with known taxonomic labels can be generated for training. The challenge is to learn a classifier $f_\theta : \mathcal{R} \rightarrow \{1, \ldots, K\}$ that generalizes well from simulated training data to accurately classify novel reads.

The specific objectives of this thesis are:

1. **Token-level representation**: Investigate whether preserving the full sequence of token embeddings from a genomic foundation model, rather than using mean-pooled summaries, improves taxonomic classification accuracy.

2. **Parameter-efficient fine-tuning**: Evaluate the effectiveness of LoRA (Low-Rank Adaptation) [13] for adapting a 498M-parameter pre-trained model to the taxonomic classification task with minimal trainable parameters.

3. **Data scaling**: Quantify the relationship between training data volume and classification performance within a fixed GFM-based architecture, and determine the data requirements for achieving competitive accuracy.

4. **Pre-training vs. tokenization**: Through controlled cross-architecture ablations holding training data and capacity comparable, isolate the contributions of (a) large-scale genomic pre-training and (b) k-mer tokenization design to taxonomic-classification performance, and identify the dominant bottleneck of the proposed approach.

5. **Multi-level and sample-level evaluation**: Assess performance across multiple metrics (micro accuracy, macro F1, top-k accuracy) at both genus and species taxonomic levels, and characterize how per-read accuracy translates to sample-level utility (relative-abundance estimation, presence/absence detection).

1.3 Contributions

The main contributions of this thesis are as follows:

1. **Systematic evaluation of GFM fine-tuning for metagenomic classification**: We

provide an end-to-end empirical study of pre-trained genomic foundation models (NT-v2, DNABERT, DNABERT-2) for short-read taxonomic classification, spanning frozen-embedding baselines, parameter-efficient fine-tuning, data scaling, RC test-time augmentation, and sample-level utility. The proposed token-level pipeline —feeding the complete token-level output of NT-v2 into a learnable attention-pooling head, fine-tuned with LoRA—preserves the positional and contextual information lost in mean-pooling approaches.

2. **Data scaling within a fixed GFM-based architecture**: Through extensive experiments spanning 500K to 50M training reads, we show that within the NT-v2 framework, training data volume is the dominant performance factor with diminishing returns: $500\text{K} \rightarrow 5\text{M}$ yields $+7.76$ pp (55.29% $\rightarrow$ 63.05%) and $5\text{M} \rightarrow 50\text{M}$ a further $+4.02$ pp (to 67.07%), while architectural and training-technique ablations yield at most ± 0.5 pp.

3. **Pre-training is beneficial under fixed 6-mer tokenization**: A controlled backbone ablation (29-layer pre-trained NT-v2 vs. a single-layer randomly initialized Transformer at the same 50M scale and identical tokenization and head) isolates a $+13.19$ pp pre-training advantage in RC-TTA accuracy (67.07% vs. 53.88%), confirming that GFM pre-training contributes meaningfully when tokenization is held constant.

4. **Tokenization is the dominant cross-architecture factor**: Holding the architecture (MetaTransformer) fixed and varying $k \in \{6, 12, 13\}$ and stride, we show that overlapping long-k tokenization is the single largest performance driver: MT 13-mer stride$=1$ trained *from scratch* on 50M reads reaches 87.42% genus Top-1, $+20.35$ pp above NT-v2 + LoRA with RC TTA (67.07%) despite no pre-training.

NT-v2's non-overlapping 6-mer tokenizer is therefore identified as the principal structural limitation of the proposed approach.

5. **Reverse-complement test-time augmentation**: We demonstrate that averaging logits from forward and RC passes at inference consistently improves accuracy by $+0.1$ to $+1.5$ pp at no additional training cost, with larger gains for pre-trained models and near-zero gain for randomly initialized architectures, supporting the interpretation that RC TTA primarily corrects pre-training-induced strand biases.

6. **Hierarchical species classification and sample-level utility**: We provide a multi-level evaluation framework for genus (120 classes) and species (1,535 classes) classification. Per-genus 50M LoRA classifiers under oracle-genus routing reach 27.8% Top-1, exceeding the flat sp_v4 oracle ceiling of 27.4% and validating per-genus specialisation as the correct hierarchical direction; the genus-routing accuracy ($\sim$66%–67%) is identified as the remaining bottleneck. At the sample level, 67% per-read accuracy translates—on species-balanced random-partition synthetic samples—to Pearson $r = 0.993$ for relative abundance and 100% sensitivity for genera at $\geq 1\%$ abundance.

1.4 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2** reviews the background and related work, covering metagenomic classification methods, genomic foundation models, parameter-efficient fine-tuning, and attention mechanisms.

- **Chapter ??** presents the proposed methodology in detail, including the model architecture, training strategy, data pipeline, and evaluation framework.

- **Chapter ??** describes the experimental setup and presents comprehensive results, including ablation studies, data scaling analysis, species-level evaluation, and computational resource assessment.

- **Chapter ??** summarizes the findings, discusses limitations, and outlines directions for future work.

Chapter 2

Background and Related Work

This chapter provides the necessary background for understanding the methods proposed in this thesis. We review existing approaches to metagenomic taxonomic classification, introduce genomic foundation models, describe parameter-efficient fine-tuning techniques, and discuss attention-based sequence aggregation mechanisms.

2.1 Metagenomic Taxonomic Classification

Metagenomic taxonomic classification aims to assign each sequencing read from an environmental sample to its source organism's position in the phylogenetic tree. We review three major paradigms for this task.

2.1.1 Alignment-Based Methods

Alignment-based methods classify reads by aligning them to reference genome databases. BLAST [4] performs local sequence alignment using heuristic seed-and-extend strategies, achieving high sensitivity but at significant computational cost (hours to days for typical metagenomic datasets). DIAMOND [5] accelerates protein-level alignments through double indexing and optimized scoring, achieving up to 20,000$\times$ speedup over BLAST while maintaining comparable sensitivity. MegaBLAST [14] optimizes nucleotide-level alignment for closely related sequences.

The primary limitation of alignment-based methods is their computational cost, which scales poorly with the size of modern metagenomic datasets containing hundreds of millions of reads.

2.1.2 k-mer-Based Methods

k-mer-based methods represent sequences as sets or distributions of fixed-length subsequences (k-mers) and use hash-table lookups for rapid classification.

Kraken2 [6] constructs a database mapping each k-mer ($k = 35$) to its lowest common ancestor (LCA) in the taxonomic tree. Classification proceeds by querying each read's k-mers against this database and assigning the read to the taxon that covers the most k-mers. Kraken2 achieves classification speeds exceeding 10^6 reads per minute, making it one of the most widely adopted tools in metagenomic analysis.

Centrifuge [7] uses a compressed FM-index to perform rapid exact matching of reads against reference genomes, achieving lower memory requirements than Kraken2 while maintaining competitive accuracy.

While k-mer methods excel in speed, they rely on exact k-mer matches and thus struggle with sequences that diverge from the reference database. They also lack the ability to capture higher-order sequence patterns beyond individual k-mers.

2.1.3 Deep Learning Methods

Deep learning methods learn discriminative representations directly from sequence data, potentially capturing complex patterns that elude k-mer matching.

DeepMicrobes [15] applies a bidirectional LSTM with attention to one-hot encoded nucleotide sequences, achieving competitive performance on species-level classification but requiring substantial training data and compute.

BERTax [16] adapts the BERT architecture to DNA taxonomic classification at superkingdom, phylum, and genus levels. Rather than relying on sequence similarity to a reference database, BERTax learns contextual k-mer representations and demonstrates generalisation to novel organisms with no close relatives in the training set. This work establishes the viability of transformer-based language models for read-level taxonomic assignment, motivating the use of larger pre-trained GFMs in this thesis.

MetaTransformer [8] is the most directly relevant prior work to this thesis. It employs a custom Transformer architecture that operates on learned 12-mer embeddings. Key design choices include:

- **12-mer tokenization**: Input reads are segmented into non-overlapping 12-mers, each mapped to a learnable embedding vector. The choice of $k = 12$ balances vocabulary size ($4^{12} \approx 16.7$M possible 12-mers, though only $\sim$1.4M appear in training data) against sequence granularity.

- **Transformer encoder**: A Transformer encoder (the best-performing configuration uses $d_{\text{model}} = 64$ with a single encoder block for $k = 13$; the authors also evaluate up to 4 blocks and $d_{\text{model}} = 128$) processes the sequence of k-mer embeddings.

- **Mean pooling**: Token representations are mean-pooled into a fixed-length vector for classification.

- **Large-scale training**: The model is trained on balanced reads simulated from 2,505 HGR-UMGS genomes using the ART Illumina simulator with a coverage factor of 3 (the exact total number of reads is not reported in the paper; we estimate $\sim$614M based on 300K training steps $\times$ batch size 2,048). The best genus-level model achieves 98.9% precision and 98.3% recall on the validation set.

A key insight from MetaTransformer is the critical role of training data volume: the authors demonstrate that performance scales substantially with data size, and that balanced sampling across taxa is essential for achieving high macro-level performance.

2.2 Genomic Foundation Models

Genomic foundation models (GFMs) are large-scale neural networks pre-trained on vast corpora of DNA sequences using self-supervised learning objectives. Inspired by the success of language models such as BERT [17] and GPT [18], GFMs treat DNA sequences as a "language" of nucleotides and learn contextual representations that capture biological structure and function.

2.2.1 Nucleotide Transformer

The Nucleotide Transformer (NT) [9] family of models, developed by InstaDeep, are among the largest and most widely evaluated GFMs. This thesis uses the `nucleotide-transformer-v2-500m-multi-species` variant, which features:

- **Architecture**: An ESM-2-based [19] Transformer encoder with 29 layers, 1,024 hidden dimensions, and 16 attention heads.
- **Tokenization**: A 6-mer tokenizer that encodes DNA sequences into non-overlapping 6-nucleotide tokens. This produces a vocabulary of $4^6 = 4,096$ possible tokens plus special tokens.

- **Pre-training**: Masked language modeling (MLM) on 850 genomes from 174 species, spanning $\sim$3.2 billion nucleotides.
- **Total parameters**: 498M parameters.

The model produces contextual embeddings for each token in the input sequence, yielding a representation tensor of shape $[\text{batch}, \text{seq\_len}, 1024]$ that captures both local sequence composition and long-range dependencies.

2.2.2 Other Genomic Foundation Models

DNABERT [10] was one of the first BERT-style models pre-trained on human genome sequences using overlapping k-mer tokenization ($k = 3, 4, 5, 6$). While effective for tasks such as promoter prediction, its relatively small scale (110M parameters) and human-genome-only pre-training limit its applicability to diverse metagenomic sequences.

DNABERT-2 [11] replaces k-mer tokenization with Byte Pair Encoding (BPE), eliminating the need to choose a fixed k and enabling more flexible sequence representation. It also adopts a multi-species pre-training strategy.

HyenaDNA [20] adopts single-nucleotide tokenization combined with sub-quadratic Hyena convolutions, enabling context windows up to 1M tokens at a fraction of the compute cost of attention-based models. Its single-character vocabulary eliminates the k design choice entirely, representing the opposite end of the tokenization design space from k-mer models.

Caduceus [21] is the first RC-equivariant bidirectional DNA language model. By constructing BiMamba layers with hard-coded reverse-complement weight sharing, Caduceus enforces strand symmetry at the architectural level rather than relying on augmentation or post-hoc inference tricks. This is directly relevant to the RC consistency analysis in Section 2.5.1; unlike Caduceus, the NT-v2 backbone does not natively enforce RC equivariance, motivating our use of RC training augmentation and RC TTA.

Evo [12] is a 7B-parameter model based on the StripedHyena architecture [22], capable of modeling DNA sequences up to 131K nucleotides. While powerful for long-range genomic modeling, its large size makes it challenging to deploy for metagenomic classification tasks involving hundreds of millions of short reads.

Across these models, a consistent theme emerges: tokenization strategy is a critical design axis. Lindsey, Fischman, et al. [23] systematically evaluate character, BPE, k-mer, Unigram, and WordPiece tokenizers across 44 genomic classification tasks and find accuracy differences of up to 5 percentage points depending on tokenizer choice, corroborating our own ablation results in Chapter **??**.

2.2.3 GFMs for Metagenomic Classification

Despite their success on single-genome tasks, GFMs have seen limited application to metagenomic read-level classification. A naïve approach extracts mean-pooled embeddings from a frozen pre-trained model and trains a downstream classifier (e.g., MLP or SVM) on these fixed representations. While simple, this approach discards the token-level structure of the representation.

Several recent works have begun to close this gap, though none directly addresses genus-level taxonomic classification of short metagenomic reads at scale. **ViraLM** [24] fine-tunes DNABERT-2 for binary virus/non-virus classification of metagenomic contigs, reporting a 22% F1 improvement on short fragments —demonstrating that GFM fine-tuning is beneficial even on compositionally diverse metagenomic sequences. **DNABERT-S** [25] adapts DNABERT-2 with contrastive learning to produce species-discriminative embeddings, achieving improvements on metagenomic binning and species clustering tasks; however, it operates at the embedding level rather than performing direct read classification. Both works fine-tune pre-existing GFMs and treat metagenomics as a secondary application; neither performs systematic tokenization ablation nor evaluates at the genus-level multi-class classification scale (120 genera, 50M reads) addressed in this thesis.

This thesis addresses the gap by proposing a token-level classification approach that preserves the full embedding sequence, fine-tunes the backbone with LoRA, and provides the first systematic comparison of pre-trained GFM fine-tuning against from-scratch Meta-Transformer training across tokenization strategies on the same large-scale dataset.

2.3 Parameter-Efficient Fine-Tuning

Fine-tuning the full parameter set of a 498M-parameter model requires substantial compute and memory, and risks catastrophic forgetting of the pre-trained representations. Parameter-efficient fine-tuning (PEFT) methods address this by adapting only a small subset of parameters while keeping the majority frozen.

2.3.1 Low-Rank Adaptation (LoRA)

LoRA [13] is based on the hypothesis that weight updates during fine-tuning lie in a low-rank subspace. For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, LoRA parameterizes the update as:

$$W = W_0 + \Delta W = W_0 + BA \tag{2.1}$$

where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with rank $r \ll \min(d, k)$. During training, only A and B are updated while W_0 remains frozen. The forward pass becomes:

$$h = W_0 x + \frac{\alpha}{r} BAx \tag{2.2}$$

where α is a scaling hyperparameter that controls the magnitude of the adaptation.

Key advantages of LoRA include:

- **Parameter efficiency**: With $r = 16$ applied to query, key, and value projections across 29 Transformer layers, only 5.54M parameters (1.11% of total) are trainable.
- **No additional inference latency**: The adapted weights $W_0 + BA$ can be merged, adding no overhead at inference time.
- **Modular**: Different LoRA adapters can be trained for different tasks and swapped at deployment time.

2.3.2 Other PEFT Methods

Adapter layers [26] insert small trainable bottleneck modules between Transformer layers. While effective, they introduce additional inference latency due to the extra forward-pass computation.

Prefix tuning [27] prepends learnable "virtual tokens" to the input sequence, influencing the attention computation without modifying model weights. It is memory-efficient but can struggle with tasks requiring fine-grained output control.

We choose LoRA for this thesis due to its combination of strong performance, parameter efficiency, and the absence of additional inference latency.

2.4 Attention Mechanisms for Sequence Aggregation

Given a sequence of token embeddings $\{h_1, h_2, \ldots, h_T\} \in \mathbb{R}^{T \times d}$, a classification task requires aggregating these into a fixed-dimensional representation. The choice of aggregation mechanism directly impacts what information is preserved.

2.4.1 Mean Pooling

The simplest approach averages all token embeddings:

$$z = \frac{1}{T} \sum_{t=1}^{T} h_t \tag{2.3}$$

Mean pooling is computationally trivial but weights all tokens equally, potentially diluting the contribution of discriminative tokens with non-informative ones.

2.4.2 Attention Pooling

Attention pooling uses learnable query vectors to compute a weighted combination of token embeddings. Given M learnable query vectors $\{q_1, \ldots, q_M\} \in \mathbb{R}^{M \times d_q}$ and linear projections $W_K, W_V \in \mathbb{R}^{d \times d_q}$:

$$K = HW_K, \quad V = HW_V \tag{2.4}$$

$$\alpha_{m,t} = \frac{\exp(q_m \cdot k_t / \sqrt{d_q})}{\sum_{t'} \exp(q_m \cdot k_{t'} / \sqrt{d_q})} \tag{2.5}$$

$$z_m = \sum_{t=1}^{T} \alpha_{m,t} v_t \tag{2.6}$$

$$z = \text{Concat}(z_1, \ldots, z_M) \in \mathbb{R}^{M \cdot d_q} \tag{2.7}$$

This mechanism enables the model to learn which positions in the DNA sequence are most informative for classification, automatically focusing on discriminative k-mers while down-weighting uninformative regions.

2.4.3 Multi-Head Attention Pooling

Extending attention pooling with multiple heads allows the model to attend to different aspects of the sequence simultaneously. Each head learns a different query, capturing diverse patterns (e.g., one head may focus on conserved marker genes while another attends to variable regions). The concatenated outputs from all heads provide a rich, multi-faceted representation for downstream classification.

2.5 Data Augmentation for DNA Sequences

2.5.1 Reverse-Complement Symmetry in DNA

DNA is a double-stranded molecule: the two antiparallel strands are linked by Watson–Crick base pairing ($A \leftrightarrow T$, $C \leftrightarrow G$). Given a sequence $s = s_1 s_2 \cdots s_L$ on the forward ($5' \to 3'$) strand, its *reverse complement* $\bar{s} = \overline{s_L} \cdots \overline{s_2 s_1}$ on the complementary strand encodes identical biological information: both correspond to the same genomic locus and therefore the same organism.

In shotgun metagenomic sequencing, a read is equally likely to originate from either strand. Consequently, a biologically correct classifier must satisfy the *strand symmetry* property:

$$f(s) = f(\bar{s}) \quad \forall\, s \tag{2.8}$$

However, standard neural network architectures do not inherently enforce this constraint. Zhou et al. [28] systematically demonstrate that deep learning models for genomics frequently violate strand symmetry, producing inconsistent predictions for a sequence and its reverse complement. Ma [29] further show that even DNA language models fine-tuned with RC data augmentation still exhibit substantial prediction inconsistency.

2.5.2 Strategies for Exploiting RC Symmetry

Three complementary strategies have been proposed to address the strand symmetry problem:

1. **Random RC training augmentation**: During training, each read is replaced with its reverse complement with probability $p = 0.5$, exposing the model to both strand orientations. While this encourages—but does not guarantee—strand-invariant representations, it effectively doubles the training data diversity at zero storage cost.

2. **RC test-time augmentation (RC TTA)**: At inference, both the forward read s and its reverse complement $\bar{s}$ are processed independently, and their output logits are

averaged before taking the argmax prediction:

$$\hat{y} = \arg\max_k \big[f_\theta(s) + f_\theta(\bar{s}) \big] \tag{2.9}$$

This "post-hoc conjoined" approach [28] enforces strand symmetry at inference time without modifying the trained model. It can be understood as a two-view ensemble where both views observe the same biological entity from different strand orientations. By the standard ensemble argument, averaging two correlated but noisy estimates of the same ground-truth label reduces prediction variance and cancels orientation-dependent errors. Zhou et al. [28] find that this approach consistently performs well across diverse genomic tasks.

3. **RC consistency training**: A regularization loss (e.g., KL divergence) explicitly penalizes divergence between $f_\theta(s)$ and $f_\theta(\bar{s})$ during training, producing a single model that is intrinsically more strand-invariant without requiring double inference at test time [29]. An alternative is to design *RC-equivariant architectures* that hard-code strand symmetry into the network structure [30], though this requires custom layers incompatible with pre-trained foundation model backbones.

In this thesis, we adopt strategies (1) and (2): RC augmentation during training and RC TTA during evaluation. We also experiment with RC consistency training (Section **??**) but find that its interaction with other regularization techniques (e.g., logit adjustment) can be detrimental, and RC TTA alone provides a reliable $+1$–1.5 percentage point improvement across all experimental settings (Section **??**).

2.6 Handling Class Imbalance

Metagenomic samples exhibit severe class imbalance: a few dominant taxa may constitute the majority of reads while rare taxa contribute only a handful. This imbalance can bias classifiers toward frequent classes, resulting in high micro accuracy but poor macro F1 due to frequent misclassification of rare taxa.

2.6.1 Loss-Level Approaches

Class-weighted cross-entropy scales the loss for each class inversely proportional to its frequency:

$$\mathcal{L} = -\sum_{k=1}^{K} w_k \, y_k \log \hat{y}_k, \quad w_k \propto 1/n_k \tag{2.10}$$

However, this approach can be unstable with mixed-precision training and large class counts [31].

Logit adjustment [32] modifies the output logits by a class-prior-dependent offset:

$$\tilde{f}_k(x) = f_k(x) + \tau \log \pi_k \tag{2.11}$$

where π_k is the prior probability of class k and τ is a temperature parameter controlling the strength of adjustment. This approach is theoretically motivated as producing Fisher-consistent estimators under class imbalance.

2.6.2 Sampling-Level Approaches

Balanced sampling constructs each training batch by sampling classes uniformly and then sampling instances within each class. This ensures that rare classes are seen as frequently as common ones during training.

Class-aware sampling (e.g., `WeightedRandomSampler`) assigns sampling weights inversely proportional to class frequency, achieving a similar effect to balanced batches within standard data loading pipelines.

2.7 Summary

This chapter has reviewed the key concepts underpinning our proposed approach: metagenomic classification methods from alignment-based to deep learning, genomic foundation models that provide rich pre-trained representations, parameter-efficient fine-tuning via LoRA, attention-based sequence aggregation, and strategies for data augmentation and class imbalance. In the next chapter, we describe how these components are integrated into a cohesive classification framework.